

Qn1

The screenshot shows the SIMATS Online Programming Lab interface. The top header is blue with a red "Secure Exit" button on the left, the title "SIMATS Online Programming Lab" in the center, and a user profile "Mohamed Affan" with ID "192512365RD" on the right. Below the header, the "PYTHON PROGRAMMING" section is active. The "PROGRAM EDITOR" tab is selected, showing a Python script. The script takes an input number 'a' and converts it to binary, octal, and hexadecimal. It also takes a binary input 'b' and converts it to decimal, octal, and hexadecimal. The "INPUT" panel on the right shows the values "67" and "111". The "OUTPUT" panel at the bottom shows the results of the conversions.

```
1 #dectoothers
2 a=int(input('Enter A Number: '))
3 binary=bin(a)
4 octal=oct(a)
5 hexadecimal =hex(a)
6 print(binary,octal,hexadecimal)
7 #bintothers
8 b=input("Enter input: ")
9 decimal=int(b,2)
10 octall=int(b,8)
11 hexa=int(b,16)
12 print(decimal,octall,hexa)
```

INPUT

67  
111

OUTPUT

Enter A Number: 0b1000011 0o103 0x43  
Enter input: 7 73 273

qn2

The screenshot shows the SIMATS Online Programming Lab interface. The top header is blue with a red "Secure Exit" button on the left, the title "SIMATS Online Programming Lab" in the center, and a user profile "Mohamed Affan" with ID "192512365RD" on the right. Below the header, the "PYTHON PROGRAMMING" section is active. The "PROGRAM EDITOR" tab is selected, showing a Python script. The script takes a complex number input 'a' and performs various operations: adding 2, multiplying by 2, and squaring it. The "INPUT" panel on the right shows the value "5+2j". The "OUTPUT" panel at the bottom shows the results of the operations.

```
1 a=complex(input("Enter input: "))
2 comp=a+2
3 print(comp)
4 comp=a*2
5 print(comp)
6 comp=a**2
7 print(comp)
```

INPUT

5+2j

OUTPUT

Enter input: (7+2j)  
(10+4j)  
(21+20j)

qn3

## PROGRAM EDITOR

Python



Run

Save

```
1 a=5/6
2 b=-10/6
3 c=-5/-8
4 d=5.8/-6.8
5 print(a,b,c,d)
```

## OUTPUT

```
0.8333333333333334 -1.6666666666666667 0.625 -0.8529411764705882
```

qn 4



```
1 a=5//6
2 b=10//6
3 c=5//8
4 d=5.8//6.8
5 print(a,b,c,d)
```

OUTPUT

```
0 1 0 0.0
```

Your INPUT goes  
here!

qn 6

# SIMATS Online Programming Lab

PYTHON PROGRAMMING



Mohamed Affan  
192512365RD

PROGRAM EDITOR

Python

Run

Save

```
1 a=5%6  
2 print(a)
```

OUTPUT

5

INPUT

Your INPUT goes  
here!

qn7

# SIMATS Online Programming Lab

PYTHON PROGRAMMING



Mohamed Affan  
192512365RD

PROGRAM EDITOR

Python

Run

Save

```
1 a=5
2 b=4
3 c=5
4 avg=(a+b+c)/3
5 print(avg)
```

OUTPUT

```
4.666666666666667
```

Your INPUT goes  
here!

qn8

# SIMATS Online Programming Lab

PYTHON PROGRAMMING



Mohamed Affan  
192512365RD

PROGRAM EDITOR

Python



Run

Save

INPUT

Your INPUT goes  
here!

```
1 a=5
2 b=4
3 t=0
4 t=a
5 a=b
6 b=t
7 print(a)
8 print(b)
```

OUTPUT

4

qn9

# SIMATS Online Programming Lab

PYTHON PROGRAMMING

Mohamed Affan  
192512365RD

PROGRAM EDITOR

Python

Run

Save

```
1 a=5
2 b=4
3 t=0
4 a,b=b,a
5 print(a)
6 print(b)
```

OUTPUT

```
4
5
```

INPUT

Your INPUT goes  
here!

qn10

## PROGRAM EDITOR

Python

Run

Save

```
1 days=450
2 months=days//30
3 rd=days%30
4 print(months)
5 print(rd)
```

## OUTPUT

```
15
0
```

## INPUT

Your INPUT goes  
here!